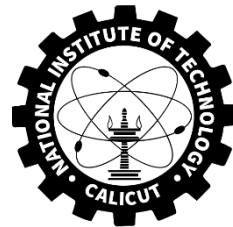


Quality, Reliability & Maintenance ME3327E



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Probability Distributions

For a discrete random variable X , which takes on the values x_1, x_2 , and soon, a probability distribution function $p(x)$ has the following properties:

$p(x_i) \geq 0$, for all i , where $p(x_i) = P(X = x_i), i = 1, 2, 3, \dots n$

$$\sum_{all\ i} p(x_i) = 1$$

When X is a continuous random variable, the probability density function is represented by $f(x)$, which has the following properties:

$f(x) \geq 0$, for all x , where $P(a \leq x \leq b) = \int_a^b f(x)dx$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Cumulative Distribution Function

The **cumulative distribution function** is usually denoted by $F(x)$ and represents the probability of the random variable X taking on a value less than or equal to x , that is,

$$F(x) = P(X \leq x)$$

For a discrete random variable, $F(x) = \sum_{all\ i} P(x_i)$ for $x_i \leq x$

If X is a continuous random variable, $F(x) = \int_{-\infty}^x f(t)dt$

Note that $F(x)$ is a non-decreasing function of x such that:

$$\lim_{x \rightarrow \infty} F(x) = 1; \quad \lim_{x \rightarrow -\infty} F(x) = 0$$

$$\frac{dF(t)}{dt} = \frac{d[\int_{-\infty}^t f(y)dy]}{dt} = f(t)$$

Expected Value

The **expected value** or *mean* of a random variable is given by

$$\mu = E(X) = \sum_{all\ i} x_i P(x_i) \quad \text{if } \mathbf{X} \text{ is discrete}$$

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \text{if } \mathbf{X} \text{ is continuous}$$

The *variance* of a random variable $\mathbf{X}$ is given by

$$Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

Exercise:

Let X denote a random variable that represents the number of defective solders in a printed circuit board. The probability distribution of the discrete random variable X is given by

x	0	1	2	3
p(x)	0.3	0.4	0.2	0.1

Evaluate mean and variance.

Expected Value

The **expected value** or **mean** of a random variable is given by

$$\mu = E(X) = \sum_{all\ i} x_i P(x_i) \quad \text{if } \mathbf{X} \text{ is discrete}$$

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \text{if } \mathbf{X} \text{ is continuous}$$

The **variance** of a random variable **X** is given by

$$Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

Exercise:

Consider a continuous random variable X representing the time taken to assemble a part. The variable X is known to be between 0 and 2 minutes, and its probability density function, $f(x)$, is given by $f(x) = \frac{x}{2}$, $0 < x \leq 2$. Evaluate mean.

Discrete Distributions

Hypergeometric Distribution: A hypergeometric distribution is useful in sampling from a finite population (or lot) without replacement (i.e., without placing the sample elements back in the population) when the items or outcomes can be categorized into one of two groups (usually called success and failure).

$$p(x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} \quad x = 0, 1, 2, \dots \min(n, D)$$

Where, D = number of nonconforming items in the population, N = size of the population, n = size of the sample, x = number of nonconforming items in the sample

$$\mu = E(X) = \frac{nD}{N}; \quad \sigma^2 = Var(X) = \frac{nD}{N} \left(1 - \frac{D}{N}\right) \left(\frac{N-n}{N-1}\right)$$

Exercise: A lot of 20 chips contains 5 nonconforming ones. If an inspector randomly samples 4 items, find the probability of 3 nonconforming chips. Evaluate mean, variance and standard deviation.

Discrete Distributions

Binomial Distribution: Consider a series of independent trials where each trial results in one of two outcomes. These outcomes are labelled as either a success or a failure. The probability p of success on any trial is assumed to be constant. Let X denote the number of successes if n such trials are conducted. Then the probability of x successes is given by

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

$$\mu = E(X) = np; \quad \sigma^2 = Var(X) = np(1 - p)$$

Exercise: A manufacturing process is estimated to produce 5% nonconforming items. If a random sample of five items is chosen, find the probability of getting two nonconforming items. Evaluate mean, variance and standard deviation.

APPENDIX A-1 Cumulative Binomial Distribution

<i>n</i>	<i>X</i>	<i>p</i> = Probability of Occurrence									
		0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
2	0	0.903	0.810	0.723	0.640	0.563	0.490	0.423	0.360	0.303	0.250
	1	0.998	0.990	0.978	0.960	0.938	0.910	0.878	0.840	0.798	0.750
3	0	0.857	0.729	0.614	0.512	0.422	0.343	0.275	0.216	0.166	0.125
	1	0.993	0.972	0.939	0.896	0.844	0.784	0.718	0.648	0.575	0.500
	2	1.000	0.999	0.997	0.992	0.984	0.973	0.957	0.936	0.909	0.875
4	0	0.815	0.656	0.522	0.410	0.316	0.240	0.179	0.130	0.092	0.063
	1	0.986	0.948	0.890	0.819	0.738	0.652	0.563	0.475	0.391	0.313
	2	1.000	0.996	0.988	0.973	0.949	0.916	0.874	0.821	0.759	0.688
	3		1.000	0.999	0.998	0.996	0.992	0.985	0.974	0.959	0.938
5	0	0.774	0.590	0.444	0.328	0.237	0.168	0.116	0.078	0.050	0.031
	1	0.977	0.919	0.835	0.737	0.633	0.528	0.428	0.337	0.256	0.188
	2	0.999	0.991	0.973	0.942	0.896	0.837	0.765	0.683	0.593	0.500
	3	1.000	1.000	0.998	0.993	0.984	0.969	0.946	0.913	0.869	0.813
	4			1.000	1.000	0.999	0.998	0.995	0.990	0.982	0.969
6	0	0.735	0.531	0.377	0.262	0.178	0.118	0.075	0.047	0.028	0.016
	1	0.967	0.886	0.776	0.655	0.534	0.420	0.319	0.233	0.164	0.109
	2	0.998	0.984	0.953	0.901	0.831	0.744	0.647	0.544	0.442	0.344
	3	1.000	0.999	0.994	0.983	0.962	0.930	0.883	0.821	0.745	0.656
	4		1.000	1.000	0.998	0.995	0.989	0.978	0.959	0.931	0.891
	5				1.000	1.000	0.999	0.998	0.996	0.992	0.984

Discrete Distributions

Poisson Distribution: A Poisson distribution is used to model the number of events that happen within a product unit (number of defective rivets in an airplane wing), space or volume (blemishes per 200 square meters of fabric), or time period (machine breakdowns per month). It is assumed that the events happen randomly and independently.

The Poisson random variable is denoted by ***X***. An observed value of ***X*** is represented by ***x***. The probability distribution (or mass) function of the number of events (*x*) is given by

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$$
$$\mu = \sigma^2 = \lambda$$

Exercise: It is estimated that the average number of surface defects in 20 m² of paper produced by a process is 3. What is the probability of finding no more than two defects in 40 m² of paper through random selection? Evaluate mean, variance and standard deviation.

APPENDIX A-2 Cumulative Poisson Distribution



X	= Mean										
	0.01	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90
0	0.990	0.951	0.905	0.819	0.741	0.670	0.607	0.549	0.497	0.449	0.407
1	1.000	0.999	0.995	0.982	0.963	0.938	0.910	0.878	0.844	0.809	0.772
2		1.000	1.000	0.999	0.996	0.992	0.986	0.977	0.966	0.953	0.937
3				1.000	1.000	0.999	0.998	0.997	0.994	0.991	0.987
4						1.000	1.000	1.000	0.999	0.999	0.998
5									1.000	1.000	1.000
	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	0.368	0.333	0.301	0.273	0.247	0.223	0.202	0.183	0.165	0.150	0.135
1	0.736	0.699	0.663	0.627	0.592	0.558	0.525	0.493	0.463	0.434	0.406
2	0.920	0.900	0.879	0.857	0.833	0.809	0.783	0.757	0.731	0.704	0.677
3	0.981	0.974	0.966	0.957	0.946	0.934	0.921	0.907	0.891	0.875	0.857
4	0.996	0.995	0.992	0.989	0.986	0.981	0.976	0.970	0.964	0.956	0.947
5	0.999	0.999	0.998	0.998	0.997	0.996	0.994	0.992	0.990	0.987	0.983
6	1.000	1.000	1.000	1.000	0.999	0.999	0.999	0.998	0.997	0.997	0.995
7					1.000	1.000	1.000	1.000	0.999	0.999	0.999
8									1.000	1.000	1.000
	2.2	2.4	2.6	2.8	3.0	3.5	4.0	4.5	5.0	5.5	6.0
0	0.111	0.091	0.074	0.061	0.050	0.030	0.018	0.011	0.007	0.004	0.002
1	0.355	0.308	0.267	0.231	0.199	0.136	0.092	0.061	0.040	0.027	0.017
2	0.623	0.570	0.518	0.469	0.423	0.321	0.238	0.174	0.125	0.088	0.062
3	0.819	0.779	0.736	0.692	0.647	0.537	0.433	0.342	0.265	0.202	0.151
4	0.928	0.904	0.877	0.848	0.815	0.725	0.629	0.532	0.440	0.358	0.285
5	0.975	0.964	0.951	0.935	0.916	0.858	0.785	0.703	0.616	0.529	0.446
6	0.993	0.988	0.983	0.976	0.966	0.935	0.889	0.831	0.762	0.686	0.606
7	0.998	0.997	0.995	0.992	0.988	0.973	0.949	0.913	0.867	0.809	0.744
8	1.000	0.999	0.999	0.998	0.996	0.990	0.979	0.960	0.932	0.894	0.847
9		1.000	1.000	0.999	0.999	0.997	0.992	0.983	0.968	0.946	0.916
10				1.000	1.000	0.999	0.997	0.993	0.986	0.975	0.957
11						1.000	0.999	0.998	0.995	0.989	0.980
12							1.000	0.999	0.998	0.996	0.991
13								1.000	0.999	0.998	0.996
14									1.000	0.999	0.999